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[24MTDS11]

FIRST SEMESTER M.Tech EXAMINATION, JANUARY - 2025

STATISTICAL METHODS AND OPTIMIZATION TECHNIQUES

Time: 3 Hours **Maximum Marks: 100**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 20 marks.
- **ii.** Missing data may be suitably assumed.

			CO's	Blooms Level	Marks
1.	**a.**	Discuss probabilistic sampling methods with appropriate examples.	**1**	**2**	**10**
	b.	A population consists of six numbers 5, 10, 14, 18, 13 and 24. Consider all possible samples of size two which can be drawn without replacement from this population. Find the following: **i.** The mean of the population. **ii.** The standard deviation of the population. **iii.** The mean of the sampling distribution of means. **iv.** The standard deviation of the sampling distribution of means.	**1**	**3**	**10**
2.	**a.**	**i.** The mean life time of a sample of 25 fluorescent light bulbs produced by a company is computed to be 1570 hours with a standard deviation of 120 hours. The company claims that the average life of the bulbs produced by the company is 1600 hours using the level of significance of 0.05. Is the claim acceptable? Given $t_{0.025,\,24} = 2.064$. **ii.** A sample of 64 students have a mean weight of 70 kgs. Can this be regarded as a sample from a population with mean weight 56 kgs and standard deviation 25 kgs? Given $Z_c = 1.96$.	**2**	**3**	**10**
	b.	**i.** The time taken by workers in performing a job by method I and method II is given below: Method I: 20, 16, 26, 27, 23, 22, - Method II: 27, 33, 42, 35, 32, 34, 38 Does the data show that the variances of time distribution from population from which these samples are drawn do not differ significantly? Given $F_{0.05}(6,\,5) = 4.95$. **ii.** The mean and standard deviation of a population are 11.795 and 14.054 respectively. What can one assert with 95% confidence about the maximum error if $\bar{x} = 11.795$ and n = 50, also construct 95% confidence interval for the true mean? Given $Z_c = 1.96$.	**2**	**3**	**10**
3.	**a.**	Discuss the properties of good estimator.	**3**	**2**	**10**
	b.	Let $f(x: \mu, \sigma)$ be given by a Gaussian distribution where x is from our set of n	**3**	**2**	**10**

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		measurements $(x_1, x_2, \dots \dots x_n)$. What is the Maximum likelihood estimation of parameters μ and σ?			
		OR			
4.	**a.**	Let $f(x, a)$ be given by a Poisson distribution where $a = m$, the mean of the Poisson distribution and x is from the set of n measurements $(x_1, x_2, \dots \dots x_n)$. What is the maximum likelihood estimation of the parameter $a = m$?	**3**	**2**	**08**
	b.	Fit an exponential curve $y = ae^{bx}$ to the following data: x : 2 4 6 8 y : 25 38 56 84	**3**	**3**	**12**
5.	**a.**	Explain different phases of Operation Research.	**4**	**2**	**08**
	b.	Solve the following LPP using Simplex method $\text{Max. } Z = 6x + 8y$ Subject to $5x + 10y \leq 60$ $4x + 4y \leq 40$ $x, y \geq 0$	**4**	**3**	**12**
		OR			
6.	**a.**	A manufacturing company makes two models A and B of a product. Each piece of Model A requires 9 labor hours for fabricating and 1 labor hour for finishing. Each piece of Model B requires 12 labor hours for fabricating and 3 labor hours for finishing. For fabricating and finishing, the maximum labor hours available are 180 and 30 respectively. The company makes a profit of Rs 8000 on each piece of model A and Rs 12000 on each piece of Model B. Formulate the linear programming problem. Apply graphical method to find (i) How many pieces of Model A and Model B should be manufactured per week to realize a maximum profit? (ii) What is the maximum profit per week?	**4**	**3**	**10**
	b.	Solve the following LPP using Big-M method $Min.\ Z = 5x + 6y$ $Subject\ to$ $2x + 5y \geq 1500$ $3x + y \geq 400$ $x, y \geq 0$	**4**	**3**	**10**
7.	**a.**	Compute an initial feasible solution to the following transportation problem using the Least cost/matrix minima method.	**5**	**3**	**06**

	D_1	D_2	D_3	D_4	*Supply*
O_1	1	2	3	4	6
O_2	4	3	2	0	8
O_3	0	2	2	1	10
Demand	4	6	8	6	24

	b.	Find initial basic feasible solution using Vogel's approximation method and find optimal solution using MODI method.	**5**	**3**	**14**

		Destination				
		P	*Q*	*R*	*S*	*Supply*
Source	*A*	21	16	25	13	11
	B	17	18	14	23	13
	C	32	17	18	41	19
	Demand	6	10	12	15	43

OR

8.	**a.**	Solve the following assignment problem whose matrix entries represent the processing times in hours using Hungarian method.	**6**	**3**	**10**

		Operator				
		1	2	3	4	5
Job	1	10	12	15	12	8
	2	7	16	14	14	11
	3	13	14	7	9	9
	4	12	10	11	13	10
	5	8	13	15	11	15

	b.	Solve optimally the following assignment problem:	**6**	**3**	**10**

		Jobs			
		I	II	III	IV
MECHANICS	A	42	35	28	21
	B	30	25	20	15
	C	30	25	20	15
	D	24	20	16	12